

WEEK 5

Simple Chessboard

Problem Statement:

Write a program that prints a simple chessboard. Input

format:

The first line contains the number of inputs T.

The lines after that contain a different value for size of the chessboard Output

format:

Print a chessboard of dimension size * size.

Print W for white spaces and B for black spaces. Sample

Input:

2

3

5

Sample Output:

WBW

BWB

WBW

WBWBW

BWBWB

WBWBW

BWBWB

WBWBW

```

1 #include<stdio.h>
2 int main()
3 {
4     int T, size;
5     scanf("%d", &T);
6     for(int t = 0; t< T; t++)
7     {
8         scanf("%d", &size);
9         for (int i = 0; i < size; i++)
10        {
11            for(int j = 0; j<size; j++)
12            {
13                if((i + j) % 2 == 0)
14                {
15                    printf("W");
16                }
17                else
18                {
19                    printf("B");
20                }
21            }
22            printf("\n");
23        }
24    }
25    return 0;
26 }

```

	Input	Expected	Got	
✓	2	WBW	WBW	✓
	3	BWB	BWB	
	5	WBW	WBW	
		WBWBW	WBWBW	
		BWBWB	BWBWB	
		WBWBW	WBWBW	
		BWBWB	BWBWB	
		WBWBW	WBWBW	

Passed all tests! ✓

PrintOurOwnChessboard

ProblemStatement:

Let's print a chessboard!

Write a program that takes input:

The first line contains T, the number of test cases

Each test case contains an integer N and also the starting character of the chessboard Output

Format

Print the chessboard as per the given examples Sample

Input:

2

2W

3B

Sample Output:

WB

BW

BWB

WBW

BWB

```
1 #include<stdio.h>
2 int main()
3 {
4     int T,N;
5     char ch;
6     scanf("%d", &T);
7     while(T-->0)
8     {
9         scanf("%d %c", &N, &ch);
10        for (int i =0; i < N; i++)
11        {
12            for (int j = 0; j < N; j++)
13            {
14                if(ch == 'W')
15                {
16                    if((i + j) %2 == 0)
17                        printf("W");
18                    else
19                        printf("B");
20                }
21                else
22                {
23                    if((i+j)%2 == 0)
24                        printf("B");
25                    else
26                        printf("W");
27                }
28            }
29            printf("\n");
30        }
31    }
32    return 0;
33 }
```

	Input	Expected	Got	
✓	2	WB	WB	✓
	2 W	BW	BW	
	3 B	BWB	BWB	
		WBW	WBW	
		BWB	BWB	

Passed all tests! ✓

PatternPrinting

ProblemStatement:

Decodethelogicandprintthepatternthatcorrespondstogiveninput. If N=

3 then pattern will be:

10203010011012

**4050809

****607

IfN=4,thenpatternwillbe:

1020304017018019020

**50607014015016

****809012013

*****10011

Constraints: $2 \leq N \leq 100$ Input

Format

FirstlinecontainsT,thenumberoftestcases,eachtestcasecontainsasingleintegerN Output

Format

FirstlineprintCase#iwhereiisthetestcasenumber,Inthesubsequentline,printthepattern

SampleInput

3

3

4

5

SampleOutput

Case #1

10203010011012

**4050809

****607

Case #2

1020304017018019020

**50607014015016

****809012013

*****10011

Case #3

102030405026027028029030

**6070809022023024025

****10011012019020021

*****13014017018

```

1  #include<stdio.h>
2  int main()
3  {
4      int t,n,x,y,z=1,i,ans,c;
5      scanf("%d", &t);
6      while(z<=t)
7      {
8          scanf("%d", &n);
9          printf("Case #%d\n",z);
10         y=1;
11         i=1;
12         c=0;
13         while(y<=n)
14         {
15             x=1;
16             ans = (n*n);
17             ans = ans - c;
18             while(x<=2*n)
19             {
20                 if(x<=n)
21                 {
22                     if(x<y)
23                         printf("***");
24                     else if (x<=n)
25                     {
26                         printf("%d",i*10);
27                         i++;
28                     }
29                 }
30                 else
31                 {
32                     if((x+y)==(2*n)+1)
33                     {

```

```

35         ans++;
36         c++;
37     }
38     else if (x+y<=(2*n)+1)
39     {
40         printf("%d", (ans+y)*10);
41         ans++;
42         c++;
43     }
44     }
45     x++;
46 }
47 y++;
48 printf("\n");
49 }
50 z++;
51 }
52 return 0;

```

	Input	Expected	Got	
✓	3	Case #1	Case #1	✓
	3	10203010011012	10203010011012	
	4	**4050809	**4050809	
	5	****607	****607	
		Case #2	Case #2	
		1020304017018019020	1020304017018019020	
		**50607014015016	**50607014015016	
		****809012013	****809012013	
		*****10011	*****10011	
		Case #3	Case #3	
		102030405026027028029030	102030405026027028029030	
		**6070809022023024025	**6070809022023024025	
		****10011012019020021	****10011012019020021	
		*****13014017018	*****13014017018	
		*****15016	*****15016	

Passed all tests! ✓

ArmstrongNumber

ProblemStatement:

Thek-digitnumberNisanArmstrongnumberifandonlyifthe k-thpowerofeachdigit sums to

N.

GivenapositiveintegerN,returntrueifandonlyifitis anArmstrongnumber. Note: 1 <=

N <= 10^8

Hint:153isa3-digitnumber,and153=1^3+5^3+3^3. Sample

Input:

153

SampleOutput:

true

SampleInput:

123

SampleOutput:

false

SampleInput:

1634

SampleOutput:

true

```
1 #include<stdio.h>
2 #include<math.h>
3 int main()
4 {
5     int N,k = 0 , sum =0 , rem;
6     scanf("%d", &N);
7     int temp1 = N, temp2 = N;
8     while (temp1 != 0)
9     {
10         temp1/=10;
11         k++;
12     }
13     while (temp2 != 0)
14     {
15         rem = temp2 % 10;
16         sum+=pow(rem,k);
17         temp2/=10;
18     }
19     if(sum== N)
20         printf("true");
21     else
22         printf("false");
23 }
```

	Input	Expected	Got	
✓	153	true	true	✓
✓	123	false	false	✓

Passed all tests! ✓

ReverseandAddUntilGetaPalindrome

ProblemStatement:

Takeanumber,reverseitandaddittotheoriginalnumberuntiltheobtainednumberis a
palindrome.

Constraints

$1 \leq \text{num} \leq 999999999$

SampleInput1

32

SampleOutput1

55

SampleInput2

789

SampleOutput2

66066

```
1 #include<stdio.h>
2 int main()
3 {
4     long long int num, sum, revnum, tempnum, tempsum;
5     scanf("%lld", &num);
6     while(1)
7     {
8         revnum = 0;
9         tempnum = num;
10        while(num)
11        {
12            revnum = revnum*10 + (num%10);
13            num = num/10;
14        }
15        sum = tempnum + revnum;
16        tempsum = sum;
17        revnum= 0;
18        while(sum)
19        {
20            revnum = revnum*10+(sum%10);
21            sum = sum/10;
22        }
23        if(tempsum == revnum)
24        {
25            break;
26        }
27        num = tempsum;
28    }
29    printf("%lld",tempsum);
30    return 0;
31 }
```


	Input	Expected	Got	
✓	32	55	55	✓
✓	789	66066	66066	✓

Passed all tests! ✓

LuckyNumber

ProblemStatement:

A number is considered lucky if it contains either 3 or 4 or 3 and 4 both in it. Write a program to print the nth lucky number. Example, 1st lucky number is 3, and 2nd lucky number is 4 and 3rd lucky number is 33 and 4th lucky number is 34 and so on. Note that 13, 40 etc., are not lucky as they have other numbers in it.

The program should accept a number 'n' as input and display the nth lucky number as output.

SampleInput1:

3

SampleOutput1:

33

```

1  #include<stdio.h>
2  int main()
3  {
4      long int i,j;
5      int rem,n,count =0,flag;
6      scanf("%d", &n);
7      for(i=1; count<=n; i++)
8      {
9          flag = 0;
10         j=i;
11         while (j>0)
12         {
13             rem = j%10;
14             if(rem == 3|| rem == 4)
15                 j = j/10;
16             else
17             {
18                 flag = 1;
19                 break;
20             }
21         }
22         if(flag == 0)
23         {
24             count++;
25             if(count == n)
26                 break;
27         }
28     }
29     printf("%ld",i);
30     return 0;
31 }
32

```

	Input	Expected	Got	
✓	34	33344	33344	✓

Passed all tests! ✓